Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

.....

.....

.....

.....[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

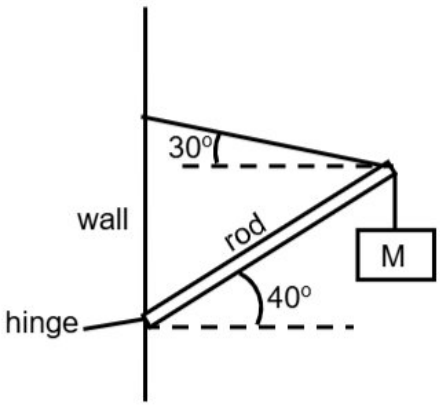


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]